

Shubham Patel

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Summary

Software Engineer with experience building backend systems, cloud-native applications, and distributed services using Python. Skilled in designing REST APIs, scalable data pipelines, and containerized systems with a focus on production deployment, reliability, and performance. Experienced in integrating AI features such as LLM-based RAG systems and computer vision models into backend architectures using FastAPI, Flask, Docker, and AWS.

Education

Humber College , <i>Graduate Certificate in Artificial Intelligence with Machine Learning</i> GPA: 92.3/100	09/2024 – 04/2025 Toronto, ON
Centennial College , <i>Graduate Certificate in Mobile Application Development</i> GPA: 4.45/4.5	09/2023 – 04/2024 Scarborough, ON
Maharashtra Institute of Technology & Art, Design & Technology , <i>Bachelor of Technology in Computer Science & Engineering</i>	07/2018 – 07/2022 Pune, India

Experience

Software Engineer - Backend & AI Systems (Client Project) , <i>Kevaras</i> - Built and deployed a real-time computer vision backend system for autonomous trash detection using Python, YOLOv11n, and OpenCV - Designed RESTful inference APIs to serve model predictions within a modular backend architecture for robotics integration - Optimized inference pipeline performance through model pruning, tuning, and system-level improvements, reducing latency by 40% - Improved model robustness by training on combined TACO and custom datasets, achieving 82% mAP@0.5 with reliable real-world performance - Collaborated with robotics engineers to integrate detection outputs into an autonomous navigation system for real-time decision-making	01/2025 – 04/2025 Toronto, ON
Software Engineer (Research & Development Project) , <i>MIT School of Engineering</i> - Designed and implemented RESTful backend APIs using Flask to serve real-time ML inference models - Built backend integration layer between Android applications and server-side APIs enabling real-time communication and data exchange - Designed end-to-end system architecture covering data flow, model serving, and deployment pipelines - Contributed to applied deep learning research and co-authored a peer-reviewed publication in computer vision systems - Contributed to production-style backend system development including API design, model serving infrastructure, and deployment workflows for ML systems.	01/2021 – 06/2022 Pune, India

Projects

- Adaptive RAG Resume Assistant** 
- Built an AI-powered backend system using FastAPI-style architecture patterns, LangChain, OpenAI APIs, and ChromaDB
 - Designed multi-step workflow orchestration for retrieval, web search (Tavily), and LLM response generation
 - Implemented structured outputs using Pydantic to enforce schema consistency in API responses
 - Built document ingestion pipeline for PDF/DOCX processing, embedding, and vector storage for semantic retrieval
 - Optimized retrieval quality using multi-query expansion and response grading, improving grounded responses by 60%
 - Deployed system with Gradio interface on Hugging Face Spaces for real-time interaction

- Customer Churn Prediction Platform** 
- Built a production-grade backend service using FastAPI, XGBoost, and MLflow for real-time predictions
 - Designed and implemented REST API endpoints for model inference and request handling
 - Containerized application using Docker and deployed on AWS ECS Fargate behind Application Load Balancer
 - Implemented CI/CD pipeline using GitHub Actions for automated testing and deployment
 - Designed cloud architecture covering model lifecycle, deployment, and scalable inference serving

- Traffic Congestion Prediction System - Distributed Data Pipeline**
- Built a distributed data processing pipeline using Apache Spark on Google Cloud Dataproc for 94M+ records
 - Designed ETL workflows for feature engineering including temporal and traffic-based transformations
 - Developed Flask-based REST API for serving predictions from trained ML models
 - Integrated ELK stack (Elasticsearch, Logstash, Kibana) for monitoring and system observability

- Designed scalable backend data processing architecture supporting batch analytics and real-time inference APIs

Skills

- **Backend & Systems:** FastAPI, Flask, Node.js, REST APIs, API Design, Microservices, System Design, Distributed Systems
- **Cloud & DevOps:** AWS (ECS, S3, EC2), Docker, CI/CD, GitHub Actions, GCP
- **Programming:** Python, SQL, JavaScript, C#
- **Data & Processing:** Pandas, NumPy, Spark, PySpark
- **AI Integration:** LLMs, RAG, LangChain, OpenAI API, Vector Databases
- **Machine Learning:** Scikit-learn, XGBoost, PyTorch, TensorFlow
- **Frontend:** React, HTML/CSS, Gradio